

Revised Generation Interconnection System Impact Study Report

For Queue Project Q16-0617

Kickapoo 100 kV
March 06, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0617. The Interconnection Customer proposes a 184.1 MW solar photovoltaic facility interconnecting at the Kickapoo 100 kV substation in the Ohio area. The request was received on May 09, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 230 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$21,074,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
STRYKER - HESPERIA 100kV	KICKAPOO - E BATAVIA 100kV	162.1	110	100.0
E BATAVIA - KICKAPOO 100kV	HESPERIA - STRYKER 100kV line	161.3	110	100.0
KICKAPOO SUBSTATION - STRYKER SUBSTATION 100kV line	E BATAVIA - KICKAPOO 100kV	152.1	117	100.0
BARK RIV - E BATAVIA 100kV	HESPERIA - STRYKER 100kV	137.4	117	100.0
LAKE MARION - SAN ANDRES 100kV	HESPERIA - STRYKER 100kV	118.4	121	100.0
BARK RIV - SAN ANDRES 100kV	HESPERIA - STRYKER 100kV	110.6	129	100.0
HESPERIA SUBSTATION - CHUB LAKE 100kV	E BATAVIA - KICKAPOO 100kV	100.1	113	69.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
STRYKER - HESPERIA 100kV	Rebuild line with higher-capacity conductor	\$4,529,000	100% (\$4,529,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
E BATAVIA - KICKAPOO 100kV	Rebuild line with higher-capacity conductor	\$4,002,000	100% (\$4,002,000)
KICKAPOO SUBSTATION - STRYKER SUBSTATION 100kV line	Rebuild line with higher-capacity conductor	\$1,231,000	100% (\$1,231,000)
BARK RIV - E BATAVIA 100kV	Rebuild line with higher-capacity conductor	\$5,052,000	100% (\$5,052,000)
LAKE MARION - SAN ANDRES 100kV	Rebuild line with higher-capacity conductor	\$1,864,000	100% (\$1,864,000)
BARK RIVER - SAN ANDRES 100kV	Rebuild line with higher-capacity conductor	\$3,154,000	100% (\$3,154,000)
HESPERIA - CHUB LAKE 100kV	Reconductor line	\$1,319,000	94% (\$1,242,000)
	Total allocated to project		\$21,074,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.